

A Reflective Journey

Entering Iowa State in August of 2020, initially as a business major, I had grave concerns: would the dining center food taste good? Does my roommate hate me? What bus route gets closest to Target? These burning questions were more important than any classwork and the pandemic happening at the time. As I've grown, left, and returned to Iowa State, and changed majors four times, I've learned that priorities will change alongside me. Now I worry about whether my cooking will taste good, whether my cat likes me, and whether I want to shop at Aldi or Walmart this week. Changing majors from business to forestry, leaving and returning to Iowa State for computer engineering, and finally graduating in cybersecurity boils down to one reason: I just had to know.

Iowa State provided me the opportunity to learn about Egyptian gods, CPU architecture, and post-World War 2 infrastructure reshaping due to the emergence of the car. However, cybersecurity classes at Iowa State have been especially eye-opening. On a global scale, every system, government, and organization is reliant on technology. The same skills I've been learning to defend against can be maliciously used to attack any part of the ever-growing Internet of Things. Understanding the topics covered in cybersecurity classes at Iowa State has strengthened my passion and fostered skills that I will use in my career. The course material has been challenging and exhilarating at the same time, and I'm excited to apply my skills in internships and develop further as an engineer post-graduation.

An influential course experience that I've had at Iowa State was taking CybE 230, cybersecurity fundamentals. This class is the introductory course that every cybersecurity student takes when majoring in cybersecurity. It teaches the basics of Linux, networking, and virtualization that industry professionals utilize every day. This course became one of my favorites because of all the opportunities and technologies that it introduces so early in the cybersecurity curriculum. The course focused on the virtualization tool Proxmox that is used on thousands of production servers to host websites and applications for millions of users. Several labs involved altering user and file permissions, managing accounts, and configuring event logging, all of which are standard practices in companies today. The final project focused on exploring an infected system that contained vulnerabilities out the wazoo. Discussing with lab mates about different bugs and the various techniques and industry tools we used to dissect the system was easily my favorite part of the class.

I enjoyed the content of CybE 230 so much that I decided to set up a home lab in my apartment. Before diving into the hobby, I had to research the hardware I needed to run virtual machines and host services on my network. To my surprise, my old laptop was a prime candidate for starting a home lab. After learning from online communities and

watching hours of YouTube tutorials, I was able to install Proxmox on my laptop, set up virtual machines, and begin hosting my own music streaming service. Setting up my private network for hosting services has been an excellent experience because it has enabled me to learn industry practices of documentation, security philosophies, and network design. As I continued expanding my home lab, I discovered that many of my classmates had also started home lab environments. Discussing new services, hardware, and lessons learned with my classmates has strengthened my understanding of networking and home lab security.

A significant experience I've had at Iowa State is working at the Adventure Program. The Adventure Program offers all things outdoors: weeklong camping excursions, indoor and outdoor climbing, and canoeing on Lake Ada Hayden. If you can do it outside, the Adventure Program does too. At the Adventure Program, I'm part of the Setting Committee, a group of five students that manages the campus climbing walls. Our committee works closely with the professional staff of Recreation Services to set standards and practices that align with both the students and the university's visions of the program. I've focused my efforts on the Setting Committee teaching new setters about climbing techniques and safety aspects of setting. Working at the Adventure Program has enabled me to explore the sport of climbing and be a part of an inviting community that shares my love for climbing and the outdoors. Working here has given me a diverse perspective on work-life balance: where do I draw the line between energy and commitment at work when it degrades my ability to enjoy my time off the clock? I believe my time working at the Adventure Program will help me balance my life and career post-graduation and beyond.

Whether it was early-morning setting routes at Lied gym or all-nighters in Coover Hall, the experiences and people I've met at Iowa State will have an impact on me forever. Iowa State will be more than an institution where I got my degree, but a place I look back on and have only great memories. While I still have plenty to learn about myself, cybersecurity, and the world, I am confident that Iowa State has prepared me to always be a student and never stop learning. Though I still worry about my cooking, whether my cat likes me, and where I'll be grocery shopping next, I am excited to move forward in my career after graduation.